

INGREDION (THAILAND) CO., LTD. — KALASIN PLANT

Project SPRINT: 18K TPA Spray Dryer & Jet Cooker

High-Availability Distributed Control System (Rockwell Automation PlantPAx 5.0 Baseline)

2 SCADA Servers

Fault-Tolerant Redundant Cluster
Primary & Secondary Hot Standby
Dell PowerEdge R550 Rackmount

1 EWS Workstation

Engineering & Diagnostics Station
Studio 5000 Logix (v33+) & FT View
Dual 27" 4K IPS Displays

3 OWS Workstations

Multi-Area Operator Clients
OWS-01 (Slurry), OWS-02 (Dryer), OWS-03
Dual 27" Industrial Monitors

PLC Redundancy

Redundant ControlLogix 5580
1756-L83E + 1756-RM2 Fiber Sync
Bumpless Switchover < 20 ms

Control & Automation System Architecture (2 Redundant Servers, 1 EWS, 3 OWS)

LEVEL 3: SERVER ROOM & DATA CENTER (FAULT-TOLERANT SCADA CLUSTER)

SCADA SERVER A (PRIMARY ACTIVE)

Dell PowerEdge R550 Rackmount (Dual Xeon, 64GB, RAID-10)
FactoryTalk View SE Server A | FactoryTalk Historian ME
FactoryTalk Directory Server | Dual 800W Redundant PSU

SYNC LINK
Heartbeat / Data

SCADA SERVER B (STANDBY REDUNDANT)

Dell PowerEdge R550 Rackmount (Dual Xeon, 64GB, RAID-10)
FactoryTalk View SE Server B (Hot Standby Failover < 1s)
Real-time Data Replica | Dual 800W Redundant PSU

CENTRAL NAS & DOMAIN CONTROLLER

AssetCentre Version Control & Automated Disaster Recovery
Windows Active Directory Domain Controller / DNS / NTP Server
Centralized Engineering Project Archive

LEVEL 2: MAIN CONTROL ROOM (MCR) CLIENT WORKSTATIONS

1x EWS (ENGINEERING)

Dell Precision Workstation (32GB, 1TB SSD)
Studio 5000 Logix Designer (v33+)
FactoryTalk View Studio Enterprise
Dual 27" 4K Displays | Stratix Mgmt

OWS-01 (OPERATOR 1)

Area 202: Slurry Prep TS-20201/02
Area 402: pH Adjust & Jet Cooker Skid
FactoryTalk View SE Client
Dual 27" Industrial Displays

OWS-02 (OPERATOR 2)

Area 602: Spray Dryer Tower CHAM-21
Triplex HP Pump PD-60202 & Air Heater BMS
Baghouse DTEX-21 & IEP Suppression
Dual 27" Industrial Displays

OWS-03 (OPERATOR 3)

Area 613: Starch Powder Silo & Packing
Area 922: SPRINT Steam Boiler (Getabec)
Area 913/930/950: LPG, WTP & Compressed Air
Dual 27" Industrial Displays

PLANT ETHERNET BACKBONE RING: Stratix 5700/5400 Managed Industrial Gigabit Switches (Redundant Fiber Ring)

REDUNDANT CONTROLLER CHASSIS (PLC-01A & PLC-01B)

- Dual Allen-Bradley 1756-A4 Chassis in Main PLC Cabinet
- Dual 1756-L83E GuardLogix 5580 Controller (10 MB Memory, 1 GHz Scan Engine)
- 1756-RM2 Redundancy Modules: High-Speed Optical Cross-Chassis Synchronization
- 1756-EN2TR Dual-Port EtherNet/IP Bridge Modules for Device Level Ring (DLR)
- Dual 1756-PA75 Redundant Power Supply Bundles with Separate AC Feeders
- Bumpless Failover Time: < 20 ms (Zero Process Disruption)

DISTRIBUTED REMOTE I/O PANELS (DLR RING TOPOLOGY)

- RIO-200 (Slurry Prep Area): 1794-AENTR Flex I/O w/ HART (1794-IB32, OB16, IF8IH)
- RIO-400 (Conditioning & Jet Cooker): 1794-AENTR Flex I/O w/ HART
- RIO-600 (Spray Dryer Tower): 1794-AENTR Flex I/O w/ HART
- RIO-MCC (Motor Control Center): Intelligent Starters & PowerFlex 755/525 VFDs
- Vendor Skid Integration: BMS Burner Panel, Getabec Boiler, IEP Explosion Suppression
- Total 472 Physical I/O Points aligned with P&ID Rev 3.5 Specification

HIGH AVAILABILITY & SYSTEM RELIABILITY HIGHLIGHTS:

- Complete Server Redundancy: 2 Dell PowerEdge R550 servers running FactoryTalk View SE active/standby pair with sub-second failover.
- Multi-Operator Coverage: 3 dedicated OWS client workstations + 1 Engineering Workstation (EWS) covering all plant areas.
- Deterministic Safety & Control: Redundant ControlLogix 5580 CPU with fiber cross-sync and Device Level Ring (DLR) network (< 3ms self-healing).
- Seamless P&ID Rev 3.5 Baseline: Harmonized with 38 P&ID drawing sheets and 472 field tags across Kalasin Plant.

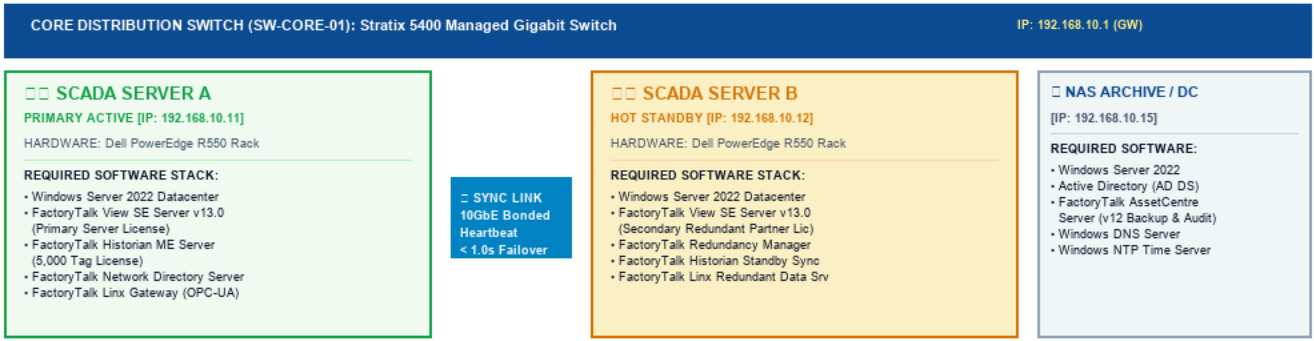
Industrial Network Topology & Required Software Architecture (CPwE / DLR)

LEVEL 3.5 : INDUSTRIAL DEMILITARIZED ZONE (IDMZ) & FIREWALL

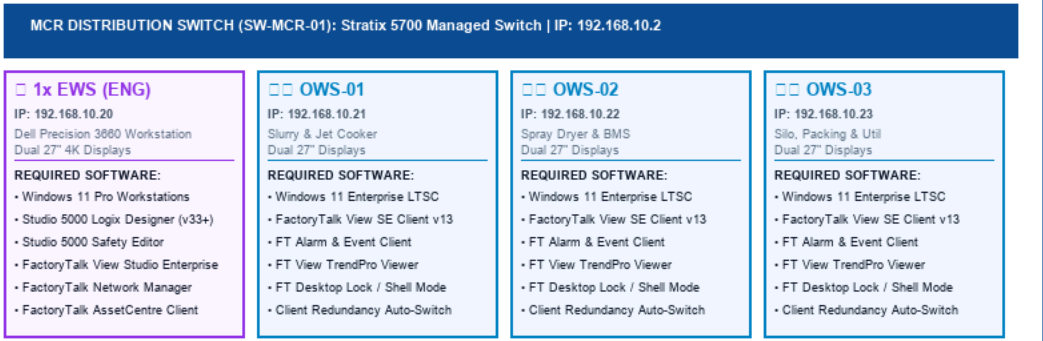
▣ FIREWALL: Stratix 5950 / Cisco Firepower

Corporate IT (10.x.x.x) <— Deep Packet Inspection (CIP/HTTPS/SSH) —> Plant OT (192.168.x.x)

LEVEL 3 : SERVER ROOM SCADA CLUSTER (VLAN 10: 192.168.10.0/24)

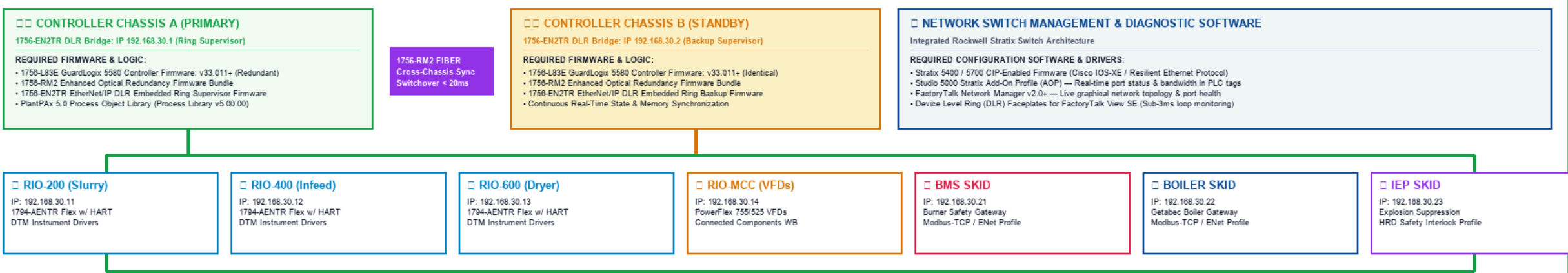


LEVEL 2 : MAIN CONTROL ROOM (MCR) CLIENTS (VLAN 10: 192.168.10.0/24)



LEVEL 1 & LEVEL 0 : CONTROLLER REDUNDANCY & FAULT-TOLERANT DEVICE LEVEL RING (DLR)

Sub-3ms Self-Healing Recovery | Zero Single Point of Failure (SPOF)



COMPLETE AUTOMATION SOFTWARE & LICENSE BILL OF MATERIALS (BOM):

SCADA Server Cluster (Primary & Secondary)

FactoryTalk View SE Server v13.0 (Redundant Pair)

9358-VWSE000LENE

2 Licenses

Active-Standby Redundant SCADA Server Engine

Industrial Data Historian

FactoryTalk Historian ME / SE (Standard Edition)

9518-HISTMESE5K

1 Server + 5K Tags

High-speed process archive & trending (100ms sample)

Engineering Workstation (EWS)

Studio 5000 Professional & FT View Studio

9324-RLD700NXENE

1 Named License

Full PLC programming, safety logic & HMI screen dev

Operator Workstations (OWS-01, 02, 03)

FactoryTalk View SE Client (Desktop Shell Mode)

9358-VWC100LENE

3 Client Licenses

Dedicated operator stations with locked OS desktop shell

Disaster Recovery & Version Control

FactoryTalk AssetCentre Server & Client

9515-ASTCNR01

1 Server License

Automated project backup, audit trail & disaster recovery

HARDWARE & SOFTWARE BILL OF MATERIALS (BOM) SPECIFICATION

Supporting 2 Redundant SCADA Servers, 1 EWS, and 3 OWS (P&ID Rev 3.5 Aligned)

SYSTEM NODE	QTY	HARDWARE SPECIFICATION	SOFTWARE & LICENSES	REDUNDANCY ROLE
SCADA Servers (Redundant)	2 Units	Dell PowerEdge R550 Rackmount (Dual Intel Xeon Silver, 64GB RAM, 4x 960GB SSD RAID-10, Dual 10GbE SFP+)	Windows Server 2022 Datacenter FactoryTalk View SE Server v13 FactoryTalk Historian ME & FTD	Primary / Standby Pair Auto-Failover (< 1.0 sec) Server Room Rack
Engineering Workstation (EWS)	1 Unit	Dell Precision 3660 Tower (Core i7, 32GB RAM, 1TB NVMe SSD, Dual 27" 4K IPS Displays)	Windows 11 Pro for Workstations Studio 5000 Logix Designer (v33+) FactoryTalk View Studio Enterprise	Dedicated Dev/Diagnostics Non-Redundant MCR Engineering Desk
Operator Workstations (OWS)	3 Units	Dell OptiPlex 7000 Series (Core i5, 16GB RAM, 512GB SSD, Dual 27" Industrial Displays per desk)	Windows 11 Enterprise LTSC FactoryTalk View SE Client v13 Auto-Login Operator Shell Mode	OWS-01: Slurry & Jet Cooker OWS-02: Spray Dryer & BMS OWS-03: Packing & Utilities
Wall Overview Displays	2 Units	50" 4K Industrial Commercial Display (24/7 Rating, HDMI Matrix Switcher)	Live Plant Overview & Master Alarm Per Ingredion Kalasin Standard Spec	MCR Wall Mounted Matrix Shared Video Feed
ControlLogix Controller Pair	2 Racks	Dual 1756-A4 Chassis, Dual 1756-L83E CPU (10MB), 1756-RM2 Redundancy, 1756-EN2TR DLR, 1756-PA75 PSU	ControlLogix Firmware v33+ Embedded GuardLogix Safety Tasks	Bumpless Optical Cross-Sync Switchover < 20 ms
Distributed Remote I/O	4 Skids	1794 Flex I/O w/ HART (1794-AENTR, IB32, OB16, IF8IH) RIO-200, RIO-400, RIO-600, RIO-MCC	FactoryTalk Network Manager Config HART Instrument Device DTMs	DLR Self-Healing Ring Recovery < 3 ms

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OPERATOR & ENGINEERING WORKSTATION ALLOCATION (MCR LAYOUT)

Functional duties and dual-display mapping for OWS-01, OWS-02, OWS-03, and EWS

OWS-01 : SLURRY PREP & JET COOKER

Primary Responsibility:

- Area 202: Re-Slurry Tanks TS-20201/02 & Agitators
- Transfer Pump PC-20201/02 Flow Management
- Area 402: pH Adjustment TS-40201/02 Skids
- Continuous Jet Cooker Chamber & Steam Infeed
- Acid, Caustic & Enzyme Dosing Loops

Screen Configuration (Dual 27" IPS):

- Screen 1: Slurry Prep & pH Adjustment Mimic
- Screen 2: Jet Cooker Real-Time Trends & Recipes

OWS-02 : SPRAY DRYER & SAFETY BMS

Primary Responsibility:

- Area 602: Spray Drying Chamber CHAM-21
- Dedert Triplex HP Pump PD-60202 (300 barg)
- Direct Gas Air Heater FH-60210A & BMS Firing
- Primary Cyclone CYCL-21 & Rotary Valves
- Exhaust Fan BF-60240 & Baghouse DTEX-21
- IEP Explosion Suppression & Chamber Deluge

Screen Configuration (Dual 27" IPS):

- Screen 1: Spray Dryer Tower & Atomizer Mimic
- Screen 2: High-Speed Trends & BMS Safety Interlocks

OWS-03 : SILO, PACKING & UTILITIES

Primary Responsibility:

- Area 613: Starch Powder Silo TS-61301 & Sifters
- Big-Bag Packer ME-61306 & Robotic Palletizer
- Area 922: SPRINT Steam Boiler (Getabec Package)
- Area 913: LPG Bullet Tank & Vaporizer Station
- Area 930: Water Treatment Plant Clarifier & Filters
- Area 950: Plant Compressed Air System (6.8 barg)

Screen Configuration (Dual 27" IPS):

- Screen 1: Packing & Bagging Lines
- Screen 2: Steam Boiler & Plant Utilities Overview

❑ 1x EWS (ENGINEERING WORKSTATION) ROLE & MCR DESK SPECIFICATION:

- Dedicated development terminal for plant automation engineers: Online ControlLogix logic modification, I/O commissioning, and HMI design.
- Equipped with Studio 5000 Logix Designer (v33+), FactoryTalk View Studio Enterprise, FactoryTalk Network Manager, and AssetCentre Client.
- High-security Windows Active Directory domain role: Restricts unauthorized operator tampering while allowing full diagnostic visibility.

HIGH AVAILABILITY, REDUNDANCY STRATEGY & FAULT TOLERANCE

Comprehensive protection against hardware, power, network, and disk drive failures

SCADA Server Redundancy (< 1.0s Failover)

- Active/Standby FactoryTalk View SE Server architecture maintains real-time tag synchronization.
- If Server A encounters hardware, OS, or network failure, Server B assumes mastership in < 1.0 second.
- Zero operator interruption: All 3 OWS and EWS automatically redirect to Server B without session restart.
- Dual 10GbE bonded NICs on Dell R550 servers eliminate network interface card dropouts.

PLC Controller Redundancy (< 20ms Bumpless)

- Dual ControlLogix 5580 chassis connected via 1756-RM2 high-speed optical fiber cross-sync modules.
- Bumpless transfer: Secondary controller assumes mastership within 1 program scan (< 20 ms).
- PID loop states, motor run statuses, and safety permissives are retained identically.
- Dual 1756-PA75 power supplies on separate AC feeds prevent power outage disruption.

Ethernet Ring & DLR Self-Healing (< 3ms)

- Device Level Ring (DLR) network provides sub-3ms recovery upon any physical fiber or cable break.
- Stratix 5700/5400 managed industrial switches provide VLAN segmentation for SCADA, I/O, and plant IT.
- Resilient Ethernet Protocol (REP) backbone ring connects Server Room, MCR, and Electrical Rooms.
- Continuous network monitoring through FactoryTalk Network Manager.

Cybersecurity & Disaster Governance (IEC 62443)

- FactoryTalk AssetCentre provides automated project version control, audit trails, and scheduled backups.
- FactoryTalk Security integrated with Active Directory domain policy ensures strict role-based access.
- Industrial firewalls isolate process OT network from enterprise IT and internet threats.
- 20% Spare I/O capacity built into RIO-200, RIO-400, and RIO-600 Flex I/O chassis for future expansion.

EXECUTIVE ENGINEERING VERIFICATION:

This automation architecture fully satisfies the engineering requirements for Ingredion Kalasin Project SPRINT, ensuring zero single point of failure, seamless operator usability across all 3 OWS, and 100% compliance with P&ID Rev 3.5.

Complete Automation Software & License Bill of Materials (BOM) — Page 7

SCADA Server Redundancy FactoryTalk View SE v13.0 1 Primary + 1 Secondary Pair	Engineering License Studio 5000 Pro + FT Studio 1 Complete EWS Suite	Operator Workstations FT View SE Client Runtime 3 Dedicated OWS Stations	Process Historian FT Historian SE Server 5,000 Tag Scalable Archive	Disaster Recovery FT AssetCentre Server v12 Automated Backup & Audit
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NO	SYSTEM NODE	SOFTWARE PRODUCT / MODULE	CATALOG / PART #	QTY	LICENSE TYPE	FUNCTIONAL DESCRIPTION & PURPOSE
01	SCADA Server A (Primary)	FactoryTalk View Site Edition Server (Unlimited Displays)	9358-VWSE000LENE	1	Perpetual / Activation	Primary active HMI data & graphics server, tag engine, alarming
02	SCADA Server B (Standby)	FactoryTalk View SE Redundant Server Partner License	9358-VWSEREDLENE	1	Perpetual / Activation	Hot-standby redundant server sync with bumpless client failover < 1s
03	Data Historian Node	FactoryTalk Historian SE Standard Edition (5,000 Tags)	9518-HISTMESE5K	1	Server Base + 5K Tags	Enterprise PI-based long-term process archiving, 100ms sample rate
04	OPC-UA Communications	FactoryTalk Linx Gateway / Data Server (DataBridge Pro)	9355-WABGWENE	2	Server License (2 Nodes)	High-performance CIP/OPC-UA server communicating with Logix CPUs
05	Engineering Station (EWS)	Studio 5000 Logix Designer Professional Edition (v33+)	9324-RLD700NXENE	1	Named User Perpetual	Full ladder, function block, structured text, SFC, and SIL2/3 logic
06	Engineering Station (EWS)	FactoryTalk View Studio Enterprise Edition	9268-FTSPENE	1	Named User Perpetual	Centralized HMI graphic design, global objects, faceplates, security
07	Engineering Station (EWS)	Studio 5000 Safety Editor & GuardLogix Safety Architect	9324-RLDSFEENE	1	Named User Add-On	Safety routine verification, checksum validation & SIL signature lock
08	Operator Stations (OWS)	FactoryTalk View SE Client Runtime License (Desktop)	9358-VWC100LENE	3	3 Device Runtime Lic	Dedicated operator client for OWS-01, OWS-02, OWS-03 (dual display)
09	Operator Stations (OWS)	FactoryTalk Desktop Lock Utility	9358-FTLOCK	3	3 Station Utility	Locks Windows OS into dedicated HMI kiosk, preventing unauthorized OS access
10	Network & Diagnostics	FactoryTalk Network Manager (Managed Switch Visibility)	9528-FTNM01	1	1 Server License	Monitors Stratix 5400/5700 switches, topology mapping & cable diagnostics
11	Audit & Disaster Recovery	FactoryTalk AssetCentre Server & Client Package	9515-ASTCNR01	1	Server + 5 Device Client	Automatic PLC program backup, change tracking, audit logs & disaster recovery
12	Server Operating Systems	Microsoft Windows Server 2022 Datacenter Edition (16-Core)	MS-WS22-DC-16C	2	OEM / Volume Lic	Hyper-V clustering, Active Directory secondary, Server A & B OS
13	Client Operating Systems	Microsoft Windows 11 Pro / Enterprise LTSC for Workstations	MS-W11-ENT-LTSC	4	OEM Digital License	Ultra-stable enterprise OS for 1 EWS and 3 OWS client stations
14	Backup & Virtualization	Veeam Backup & Replication Enterprise Edition	V-VBR-ENT-01	1	Host Socket / VM Lic	Full bare-metal backup & instant VM restore for SCADA servers and NAS

LICENSING NOTES & LIFECYCLE COMPLIANCE:

- All Rockwell Automation software licenses are managed via FactoryTalk Activation Manager (FTAM) hosted on Primary Server A (with Server B secondary sync).
- Software stack includes 1-year TechConnect Comprehensive Support 24x7. Fully compliant with PlantPAx 5.0 process library & ISA-88 batch standards.

APPROVED FOR PROCUREMENT
AEC Engineering / Ingredion Kalasin